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CLASSIFICATION OF IMAGERY MOTOR EEG DATA WITH WAVELET DENOISING AND FEATURES SELECTION

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Abstract:

Classification of imagery motor tasks is the main challenge of analysing electroencephalography (EEG) data from a brain-computer interfaces (BCI) system. The noise and artifacts recorded by BCI system frequently corrupt imagery motor EEG data and reduce classification accuracy. In this paper, wavelet denoising algorithm is proposed to reduce noise from motor imagery EEG data and a power spectral density (PSD) feature selection method is used to improve classification accuracy. Experimental results show that the classification accuracy of the proposed method is significantly improved compared to the same PSD feature selection method without wavelet denoising. This result also certainly indicated that wavelet denoising algorithm successfully purified motor imagery EEG data and made classifying features more prominent.

Keywords:

Brain–computer interfaces; Electroencephalogram (EEG); Motor imagery classification; Wavelet denoising; Power spectral density (PSD) Estimation

1. Introduction

A brain computer interface (BCI) is a communication system which carries the brain's messages to a computer. The BCI system recognizes people thoughts by extracting certain information from brain signals and uses the provided information to operate external devices [1]. Electroencephalography (EEG) measures cerebral neuron electrical signals and tracks brain activity in real time. An EEG possesses several advantages, such as non-invasive, high temporal resolution, relatively low cost regarding data collection and high portability. Based on these characteristics, EEG techniques are widely applied in a BCI system.

Analyzing motor imagery (MI) activity aims to aid those physical disabled or amyotrophic. Most research of MI activity normally includes left or right hand movement and tongue or feet movement [2]. During the MI tasks, the subject needs to commit a mental imagination of the specific movements, creating brain signals which will be recorded as EEG data. The goal of MI research is to classify MI EEG data by recognizing certain patterns. MI research shows that MI often causes an event-related desynchronization (ERD)/event-related synchronization (ERS) of oscillations in special frequency bands, specifically alpha band (8-13 Hz) and beta band (14-30 Hz). Due to the points previously referred to [3], many researchers started to classify MI EEG data with quantifying ERD/ERS oscillations. Additionally, Because of the obvious time and frequency domain features of MI EEG data, other different algorithms have been proposed, such as adaptive autoregressive (AAR) coefficients[4], Hjorth parameters [5], common spatial patterns (CSP) [6], hidden Markov models (HMMs) [7], neural network [8], and power spectral density (PSD) [9].

Unfortunately, most traditional MI EEG data analysis methods have neglected the importance of signal denoising before implementing algorithms. Since an EEG signal is a weak, low SNR signal, it is extremely sensitive to noise. Specifically, a right hand MI task could be classified as a right hand MI task due to noise interference. In order to cope with this problem, there must be a denoising step for MI EEG data classification, such as a wavelet algorithm which we have proposed in this paper. In light of ERD/ERS research [10], PSD is a distinct feature for discriminating MI EEG data. Moreover, the PSD feature extraction method has other advantages: algorithm high maturity, low consumption, and high operation speed.

In this paper, we introduce a new feature extraction method including wavelet denoising and the PSD feature selection of channel pairs. Experimental data is classified with a support vector machine (SVM) method. To carry out this proposed method, we use a PhysioNet EEG motor movement/imagery dataset [11], [12].

2. Method

2.1. Wavelet denoising

Based on a wavelet transform algorithm, useful signals

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normally have a larger wavelet coefficient while noise has a smaller wavelet coefficient. Donoho and Johnstone proposed a method: they created a threshold, set the wavelet coefficients to zero if they were smaller than threshold, and, as the inverse of wavelet transformed, received a denoised signal [13].Donoho and Johnstone also gave a universal threshold δ defined as

$$\delta=\sigma\sqrt{2\ln N} \tag{1}$$

with $\sigma=M/0.6745$, where σ is a rough estimate of noise level, and M is the median absolute deviation (MAD) of detail coefficients at highest resolution level.

However, this universal threshold critically shrank wavelet coefficients; consequently some useful data of higher frequencies were removed as noise. Thus, the problem of threshold choice needs to resolve for application of wavelet denoising.

There are two threshold functions: hard-threshold and soft-threshold [14]. The hard-threshold function might make data lose some useful information, therefore, we must adopt the soft-threshold denoising algorithm.

For noisy $x(i)$ signal

$$x(i)=s(i)+z(i) \quad i=1,...,N$$

Where $z(i)$ is a Gaussian white noise, N is the length of the signal. The soft-threshold denoising algorithm implements as below steps:

1. Wavelet transform $x(i)$, and get all wavelet coefficient $W_{jk}(j=1,...,J;k=1,2,...,n_j)$;
2. Choose a threshold T_{thr} according to threshold selection rule.
3. Calculate the new wavelet coefficient W_{jk} with soft-threshold function:

$$W_{jk}=\begin{cases}\operatorname{sgn}(W_{jk})(\left|W_{jk}\right|-T_{thr}) & \left|W_{jk}\right|>T_{thr}\\ 0 & otherwise\end{cases} \tag{2}$$

4. Inverse with W_{jk} to obtain the denoised signal $s(i)$.

2.2. Data acquisition and experimental paradigm

The PhysioNet EEG motor movement/imagery dataset is used in this work. The EEG data is recorded from 64 electrodes as per the international 10-10 system, and the numbers below each electrode name indicate the order in which they appear in the records. This dataset consists of 109 subjects, who performed different motor/imagery tasks while 64-channel EEG scans were recorded using the BCI2000 system, as shown in Figure 1. We randomly chose 9 subjects for the experiment. Each subject performed 14 experimental runs, including 3 sessions, which involved imagining themselves opening and closing their left or right fist, and every session is 2 minutes. Every session included 30 trials, including rests as well as the left and right fist MI tasks, and every trial is 4 seconds. The sample rate was at 160 samples per second. In total, there were 15 MI tasks per session. Thus, the dataset for each subject contained a total of 45 trials.

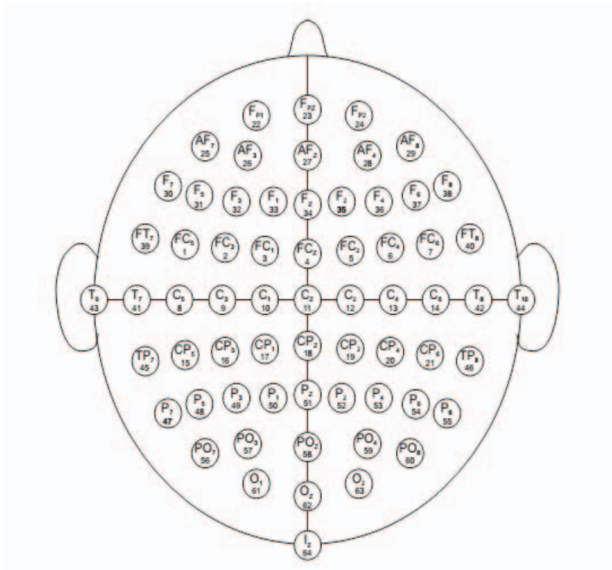

FIGURE 1. 64-channel EEG

2.3. Feature extraction

According to fMRI observation, Colors which present ERD/ERS oscillations change over the primary sensorimotor area during MI tasks [15]. Furthermore, the supplementary motor area (SMA) is also vital part to indicate the preparation and organization of voluntary MI. For some subjects, obvious similar ERD/ERS phenomena in primary sensorimotor area also can be observed in SMA area [16]. Base on above, in our experiment, we only select several channels which are located over the primary sensorimotor and SMA regions. Therefore, channels C1to C6 and CP1 to CP6 are utilized as raw MI EEG input signals.

The next step is that raw MI EEG signal of every channel is fed to wavelet soft-threshold denoise. On account of ERD/ERS phenomena, which can be observed at special frequency bands, most are alpha band (8-13 Hz) and beta band (14-30 Hz), after all channels' MI EEG data is denoised, then they are sent to filter to 8-13 Hz and 14-30 Hz by a fourth-order Butterworth bandpass filter. Here, the length of 1 trial is assumed as N , therefore every trial is 12 (channel)$\times$2 (frequency band) N .

Once denoised MI EEG signal is filtered, PSD of every frequency band signal is calculated. In this paper, we use Welch power spectral density estimation method because of convenience and speed of calculations [17].

The Welch method is an improved periodogram, whose variance performance is better than the original periodogram. The signal is separated to segments; the length of every segment is M. Then, the periodogram of one segment is:

$$J_p(w) = \frac{1}{MU}\left|\sum_{n=0}^{M-1} x_p(n)w(n)e^{-jwn}\right|^2 \qquad (3)$$

Where $U(w) = \frac{1}{M}\sum_{n=0}^{M-1} w^2(n)$, $w(n)$ is window function. Consequently, the signal $x(n)$ has PSD estimation which is:

$$B_x(w) = \frac{1}{P}\sum_{p=0}^{P} J_p(w) \qquad (4)$$

To estimate PSD, a hamming window is adopted. Because of the BCI system's recording limitations, a subject may not hold similar amplitude while the same MI task is carried out at different times. Considering this problem, the estimated PSD needs to normalize as below:

$$\overrightarrow{B_x(w)} = \frac{B_x(w) - \frac{1}{W}\sum_{w=1}^{W} B_x(w)}{\max\left(B_x(w)\right) - \min\left(B_x(w)\right)} \qquad (5)$$

Where W is length of $B_x(w)$, and $\overrightarrow{B_x(w)}$ is normalized PSD feature.

Most of ERD/ERS research shows that different subjects present ERD- /ERS-related MI EEG PSD in different channel pairs, so finding out the best channel pair became the key to increase classification accuracy.

2 sessions (30 trials) are set as the training data, while 1 session (15 trials) is testing data of one subject. We use a support vector machine (SVM) as a classifier.

In the training phase, the extracted feature of one trial is 12$\times$2W. 30 trials are separated again into 2 fold which are 20 training data and 10 testing data for validating the best channel pair. Every time, one channel pair features is trained and tested, then we can collect all the channel pairs' EEG data classified results. Finally, the best channel pair with the highest accuracy can be found. In the evaluation phase, only features of the best channel pair are selected for the 30 training data. By using SVM classifier, we can get the optimal classification surface to evaluate the testing data with the best channel pair features.

For illustrating the importance of wavelet denoising, we compare the denoised MI EEG data classification with the non-denoised MI EEG data classification.

3. Results and conclusions

This paper illustrated the performance of the wavelet denoising and feature selection method on a PhysioNet EEG motor movement/imagery dataset. A certain channel pair was selected as the best feature for MI EEG data classification. Table 1 shows the best channel pair for each subject.

TABLE 1. The best channel pair

Subject	The best channel pair
S01	C5-C6
S02	CP5-CP6
S03	CP5-CP6
S04	C3-C4
S05	CP1-CP2
S06	CP5-CP6
S07	C3-C4
S08	C1-C2
S09	C3-C4

TABLE 2. Classification results

Subject	Denoised	Non-denoised
S01	0.67	0.67
S02	0.67	0.67
S03	**0.42**	0.41
S04	0.73	0.8
S05	**0.6**	0.35
S06	**0.73**	0.47
S07	**0.86**	0.53
S08	**0.67**	0.47
S09	**0.67**	0.6
Average	**0.65**	0.55

Denoised MI EEG data was then compared with non-denoised MI EEG data. In the case of non-denoised MI EEG data, we directly filtered the raw MI EEG data to the alpha and beta band, then extracted the PSD feature for both frequency bands of all 12 channels. We also selected the best channel pair with the highest accuracy on the training data and got the non-denoised MI EEG data evaluation results.

As Table 1 shows, the best channel pair for most subjects are CP5-CP6 and C3-C4, which means that CP5-CP6 and C3-C4 have more significant ERD-/ERS-related PSD feature for MI classification. Moreover, this result also explained why most research used C3-C4 channel pairs as feature extraction channels.

The comparison between the denoised method and the non-denoised method is shown in Table 2. We observed that the method proposed in this paper provides extremely higher accuracy than the non-denoised method, and the average classification accuracy of all nine subjects is 10% higher than non-denoised method. These results conclude that the denoised method extracted more pure EEG data for MI tasks and made PSD features more outstanding.

However, for subject S01 and S02, the results between the two methods were same. The reasons may be that there were less noises interrupting the BCI system recording for subject S01 and S02; there were special noises of them which are difficult to remove with wavelet denoising algorithm. Subject S04 had contrary results, where the non-denoised method performed better than denoised method. This may be due to the fact that the wavelet denoising algorithm may have shrank some of the useful features of this subject.

A total analysis from Table 2 reflects that the wavelet denoising algorithm could be more effective in processing EEG data and classifying MI tasks. The reason why wavelet denoising could lead to better performance is that the purified MI EEG data could be extracted with more distinct features through variable feature extraction algorithms, especially for MI tasks where noises are easier to corrupted ERD-/ERS-related information.

This paper introduced a new feature extraction method to apply on MI EEG data, a method which combined a wavelet denoising algorithm and a PSD feature extraction method. The results showed that, this new method made classification accuracy higher; it extracted unique features for every subject, through the process of which we succeeded in achieving even more precise results. For future research, we can use more efficient feature extraction methods to replace the PSD method, such as CSP and ICA. Finally, the proposed denoised and feature selection method can get remarkable MI task classification results.

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